accelerate training and improve performance, especially for smaller datasets.

3. **Batch Normalization**:
    - Adding batch normalization layers can stabilize training by normalizing inputs to each layer.
4. **Advanced Architectures**:
    - Explore deeper architectures with additional convolutional and pooling layers for more complex tasks.

CNNs are a powerful tool for image classification and other computer vision tasks. By following this guide, you can build, train, and evaluate a simple CNN using TensorFlow and Keras. With practice and experimentation, you can extend this basic model to tackle more challenging image recognition problems.

Real-World Example: Handwritten Digit Recognition (MNIST Dataset)

The MNIST dataset is one of the most well-known benchmarks in the field of deep learning and computer vision. It contains 70,000 grayscale images of handwritten digits, each representing a digit from 0 to 9. The dataset is widely used for demonstrating the power of neural networks, particularly convolutional neural networks (CNNs), in solving image classification tasks.